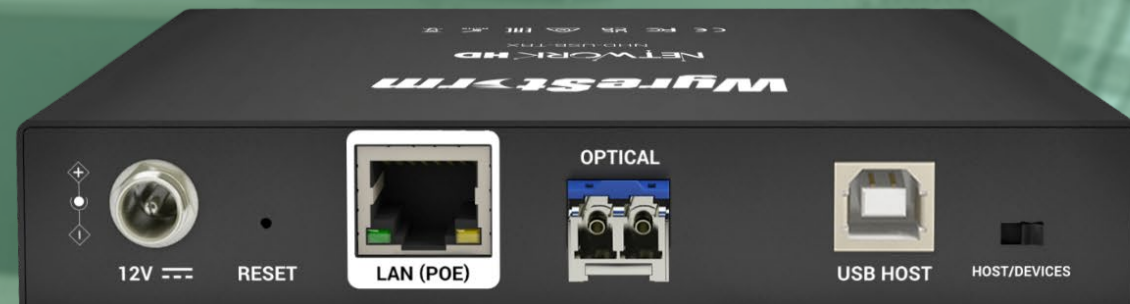


Product Guide

NHD-USB-TRX

USB 2.0 over IP



**Award-Winning Audio-Visual
Solutions Manufacturer**

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Product Overview

Seamless Interoperability with NHD-500

- Fully compatible with the **NHD-500 series encoders and decoders**, leveraging their built-in USB functionality.
- Centralized control and management through the **NHD-CTL-PRO controller** for system-wide integration.



USB2.0 over IP

- Delivers **USB 2.0 over IP** functionality, enabling **one-to-one** and **one-to-many** connections.
- Supports a wide variety of USB 2.0 input devices, ensuring adaptability for diverse application scenarios.

USB3.2 Gen2 Local Hub (HOST Mode)

- Features a **USB 3.2 Gen2 local USB hub** with up to **10Gbps** data transfer speed for devices requiring fast data rates.

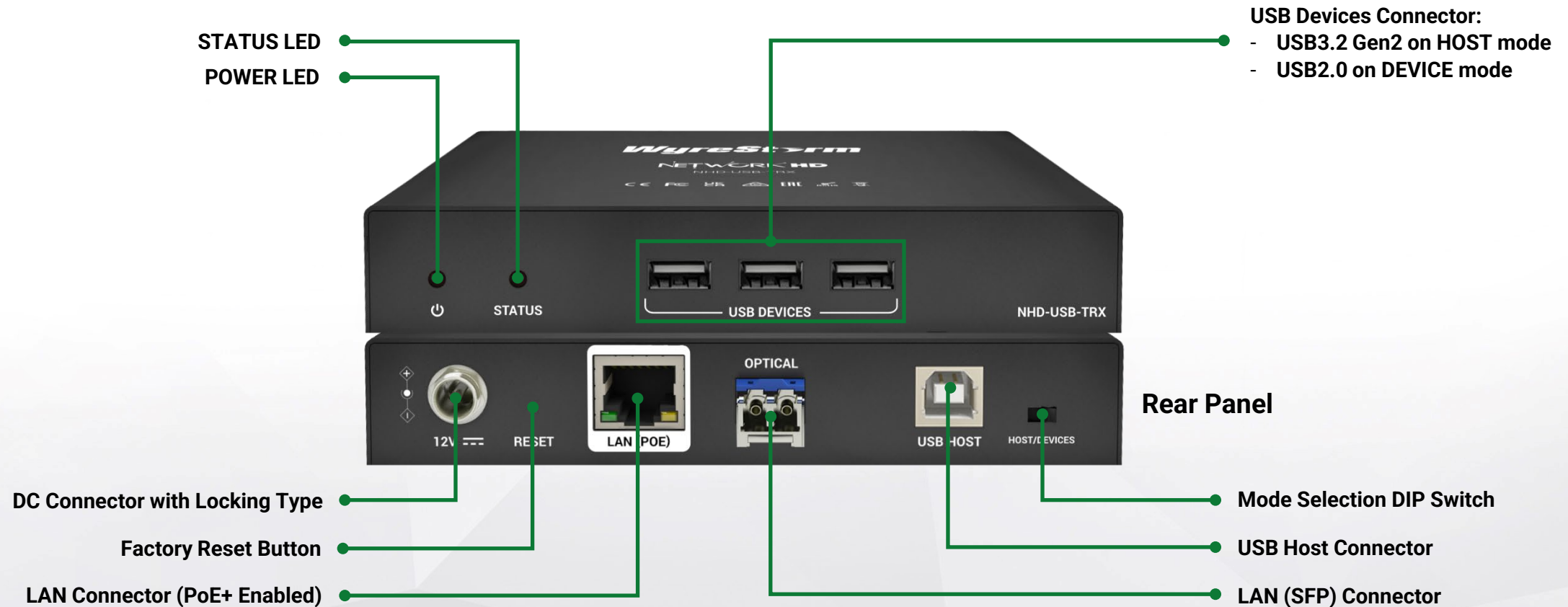
PoE+ Technology

- Integrated **PoE+ (Power over Ethernet)** allows a single cable for both **network connectivity** and **power delivery**.
- Provides sufficient power for high-current USB devices, eliminating the need for additional power adapters.

Single SKU for HOST and DEVICES Mode

- Easily switch between **HOST mode** and **DEVICE mode** via **DIP switch** or **NHD-CTL-PRO controller**.
- Simplified SKU management for greater deployment flexibility.

Interface of the Product



Specifications & Packing List

Main Specifications

USB Host	1x USB3.2 Gen2 Type-B, 10Gbps
USB Device	- HOST mode: 3x USB3.2 Gen 2 Type-A, 10Gbps - Device mode: 3x USB2.0 Type-A, 480Mbps
LAN	- 1x RJ45 (PoE+) - 1x SFP
Control	- DIP Switch - NHD-CTL-PRO - Console API
Power Supply	12V 2A DC

What's in the Box?

- 1x NHD-USB-TRX
- 2x Plastic Mounting Brackets
- 1x Quick Start Guide

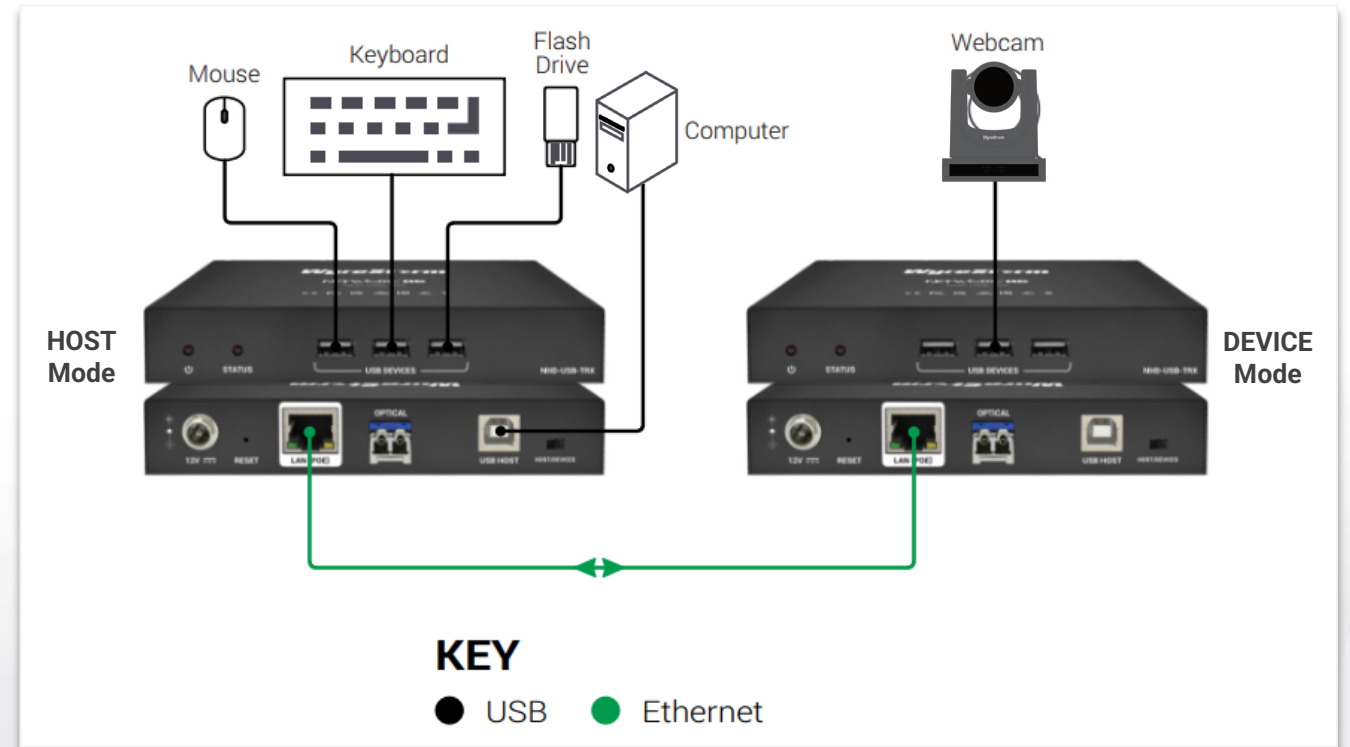
NOT in the Box

- Power Supply
- 1G DLC SFP Module

Typical Connection & Applications

Application Scenario 1: Point to Point

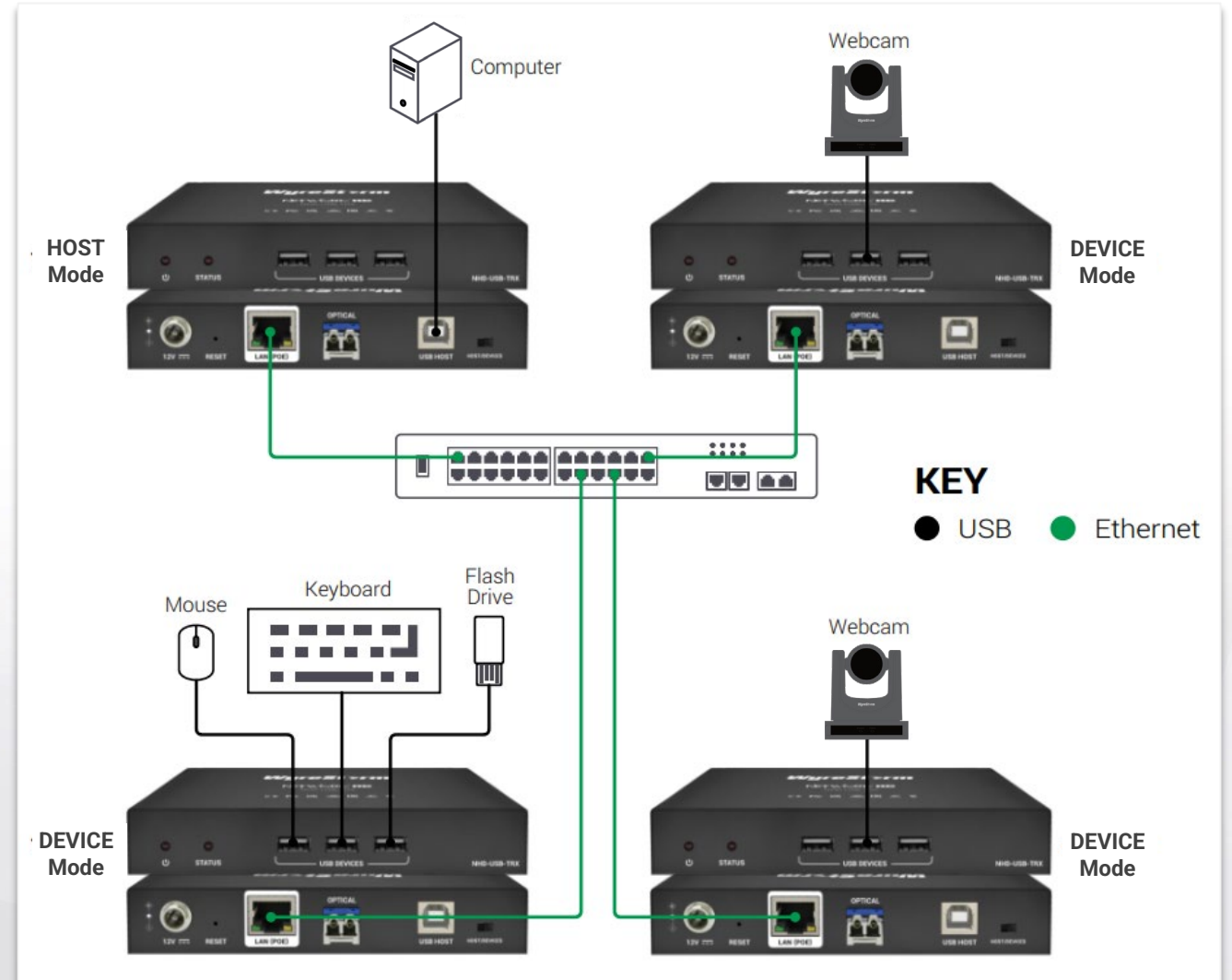
- No **NHD-CTL-PRO** required for operation
- **USB 3.2 Gen2** (up to 10Gbps) for local high-speed connections.
- **USB2.0** for remote device connectivity.



Typical Connection & Applications

Application Scenario 2: Point to Multipoint

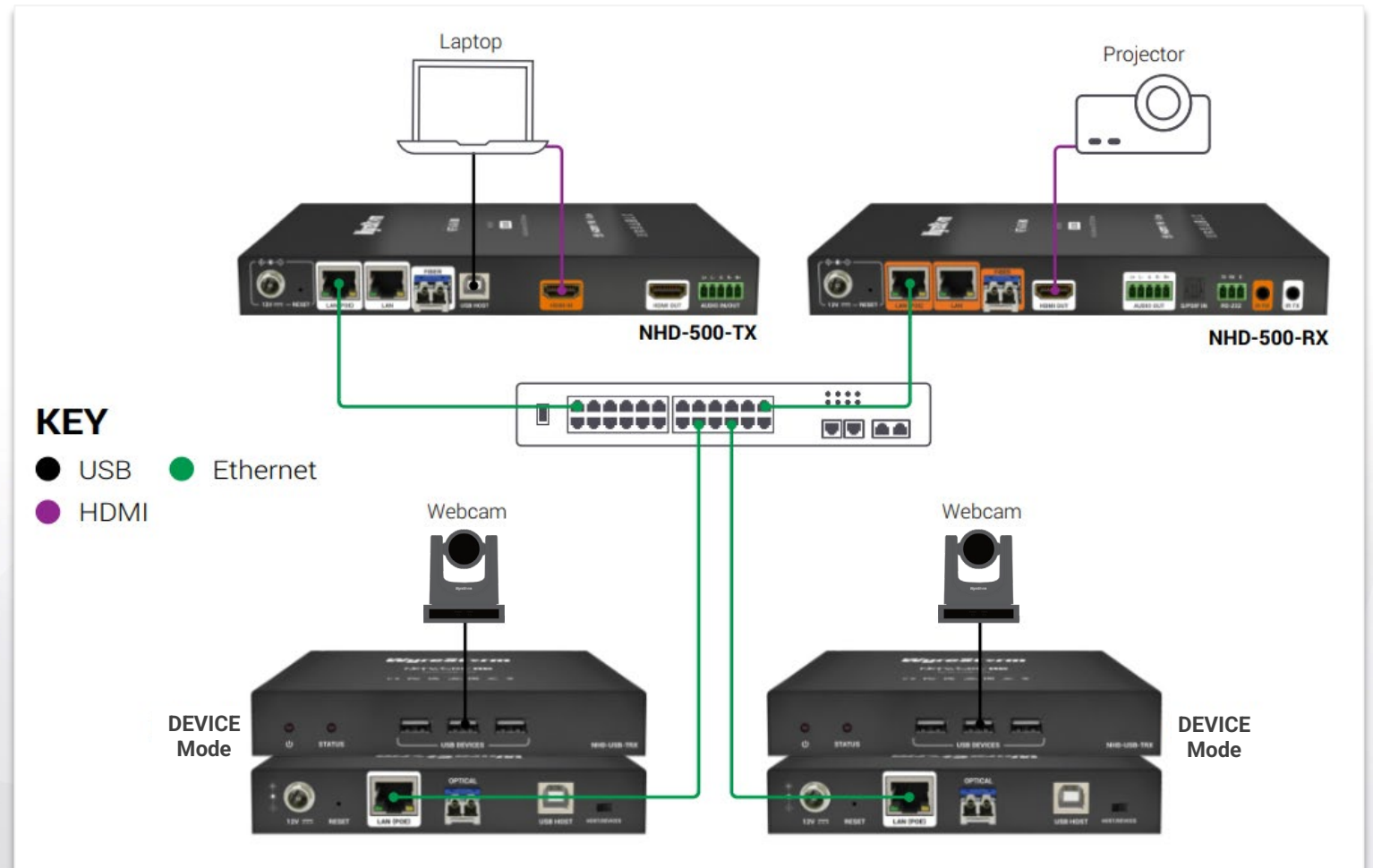
- One host connecting to multiple devices in different locations.
- Managed by **NHD-CTL-PRO** for centralized control.
- **USB 3.2 Gen2** (up to 10Gbps) for local high-speed connections.
- **USB2.0** for remote device connectivity.



Typical Connection & Applications

Application Scenario 3: Interoperable with NHD-500

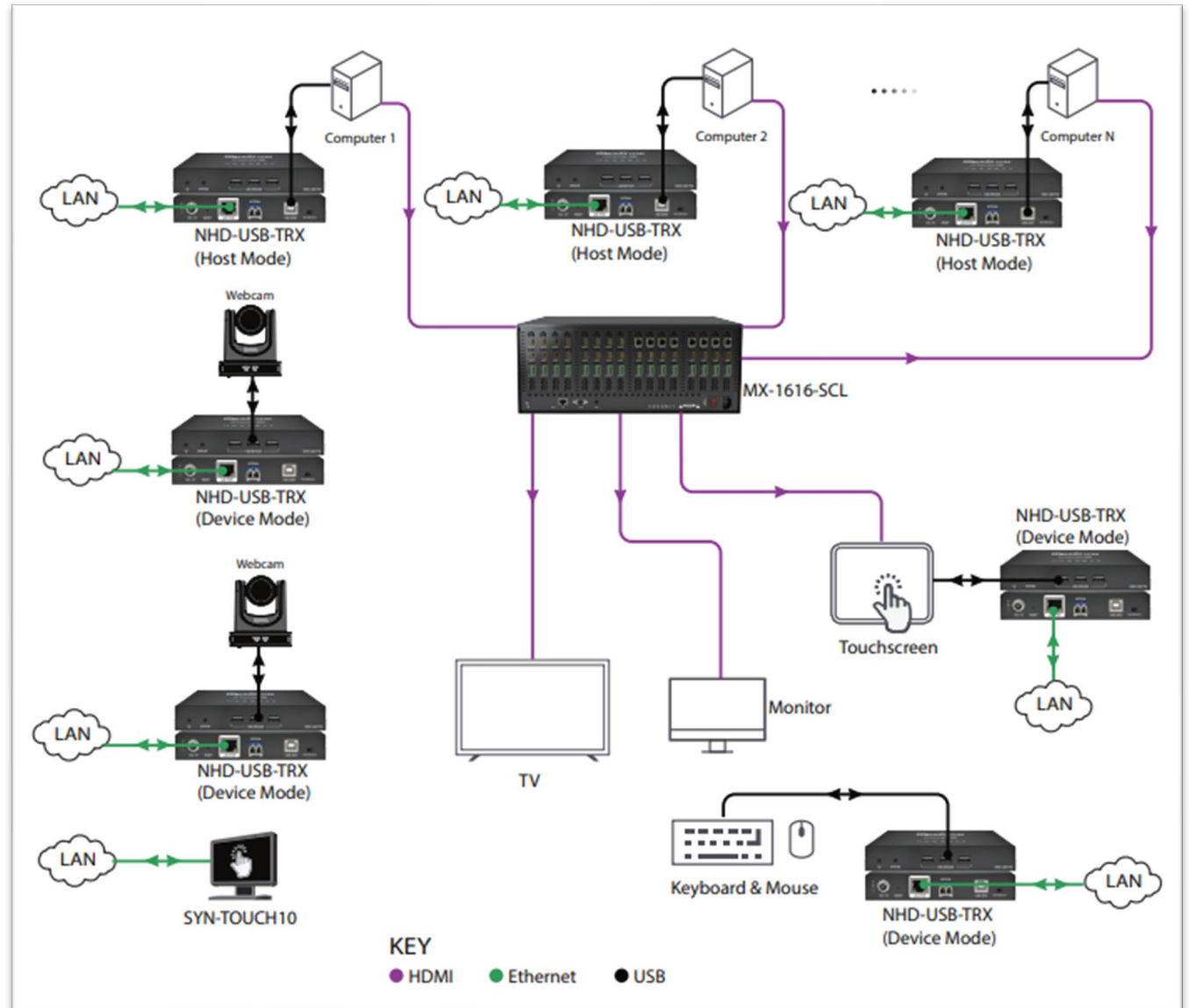
- One host connecting to multiple devices in different locations.
- Acts as a **USB-only extension** for the NHD-500 ecosystem.
- Fully managed by **NHD-CTL-PRO** for seamless integration.
- **USB2.0** for remote device connectivity.



Typical Connection & Applications

Application Scenario 4: USB over IP Solution with Matrix

- Each host can connect to multiple devices in different locations.
- **USB over IP** solution offers more wiring flexibility and possibilities in many scenarios.
- The routing relationship between hosts and devices can be flexibly adjusted according to the scenario.
- Fully managed by **SYN-TOUCH10** for seamless integration.
- **USB2.0** for remote device connectivity.



Comparison with Other Competitors

	WyreStorm NHD-USB-TRX	Icron (ADI) USB 2.0 Ranger 2304PoE
USB 2.0 High Speed	480Mbps	480Mbps
USB Local Hub on TX	USB3.2 Gen 2 10Gbps	USB2.0 480Mbps
USB Downstream Ports	3	4
Max. Power Current on USB Ports	3.0A total shared	1.5A at each port, 3.5A total
Max. Devices Quantity	7	30
Max. Endpoints Quantity	21	Unlimited
Measured Camera Bandwidth	160Mbps	160Mbps
USB Hub/Tier Usage	Two Hubs total, one on HOST end, one on DEVICE end	Two Hubs total, one on LEX, one on REX
Working with NHD-500 family	Yes	No
Transceiver	Yes	No

Our Advantages

The NHD-USB-TRX delivers a flexible, scalable, and cost-effective solution, ideal for conference rooms, classrooms, and professional environments, ensuring seamless connectivity and efficient performance for all your presentation and conferencing needs.

Our key advantages include:

- **USB2.0 Over IP:** Supports up to 7 USB devices and 21 endpoints for flexible, scalable one-to-many setups.
- **NHD-500 Compatibility:** Seamlessly integrates with the NHD-500 family for smooth system management and easy setup.
- **Transceiver Flexibility:** One SKU for both Transmitter and Receiver, streamlining inventory management and system configuration.
- **USB Camera Support:** Ideal for video conferencing and interactive presentations, with native support for USB cameras.
- **Cost-Effective:** Offers a powerful, affordable solution for professional environments.



Quick Installation Guide

To install the presentation device, it simply takes 4 steps below:

Configuring HOST or DEVICE Mode

- Use the **DIP switch** to set the **NHD-USB-TRX** to either **HOST** or **DEVICE** mode.

Connect and Power Up

- Connect USB cables, Ethernet cable, and power up the device using either PoE+ or a power adapter.

Configure Routing via NHD-CTL-PRO

- Log in to the NHD-CTL-PRO controller.
- Set up the routing between TX (transmitter) and RX (receiver).
- Optionally, configure the device to work seamlessly with the NHD-500 family endpoints.

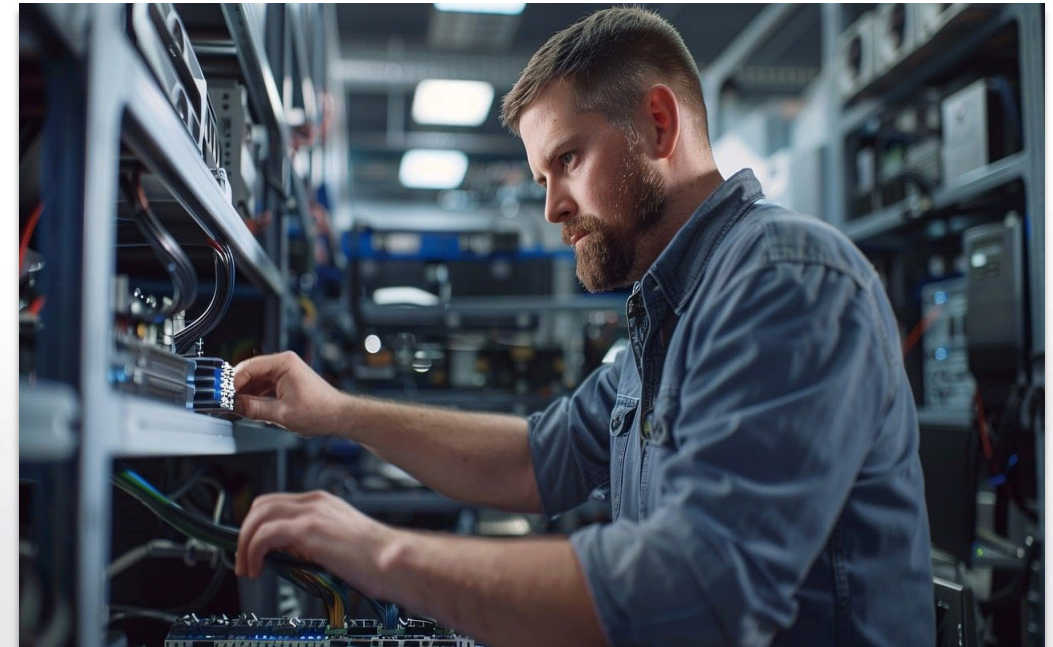
IMPORTANT

PoE+ Compatibility

- Using a PoE+ (802.3at) capability network switch instead of a PoE (802.3af) only for getting full power for USB devices.

Power Supply Option

- When not using PoE+, ensure you use a **compatible power adapter** capable of supplying at least 2A current to power the device effectively



Introduction of Web Configuration Interface

The **Web Configuration Interface** on **NHD-CTL-PRO** provides an intuitive platform for managing devices and streamlining workflows.

Key functionalities include:

Firmware Management

- Perform **simultaneous firmware upgrades** for one or multiple **NHD-USB-TRX** devices, saving time and effort.

Routing Management

- Configure **routing between NHD-USB-TRX** devices on the HOST and DEVICE ends.
- Seamlessly integrate and manage routes involving **NHD-500 Encoders or Decoders**.

Working Mode Configuration

- Dynamically switch **USB-TRX modes** directly via the **NHD-CTL-PRO** interface, even during runtime.



Final Thoughts

Q & A Session

Got questions? Let's dive into a lively Q&A session!
Your curiosity fuels the conversation, so don't hesitate to ask away.

